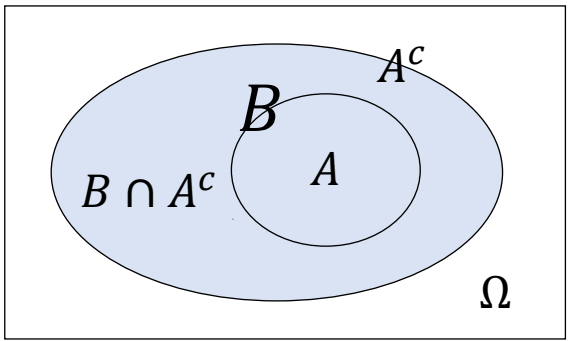
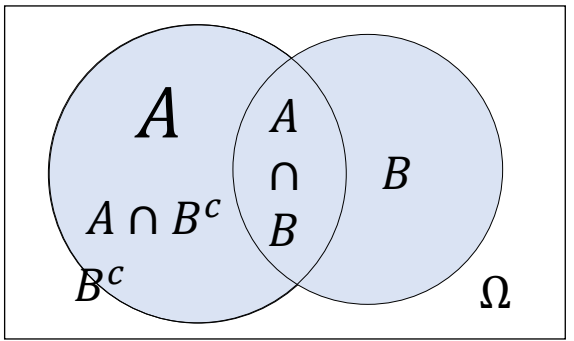
$$A \subset B \Rightarrow B = A \cup (B \cap A^c)$$

$$A \cap (B \cap A^c) = \emptyset$$



$$A = (A \cap B) \cup (A \cap B^c)$$

$$(A \cap B) \cap (A \cap B^c) = \emptyset$$



$$A \cup B = B \cup (A \cap B^c)$$

$$B \cap (A \cap B^c) = \emptyset$$

